Ahmed Hussien

ahmedhussien783@gmail.com (+2)01099668644 LinkedIn GitHub

Professional Experience

Software Engineer – Visualization and Debug Team Mar 2023 – Present
Siemens EDA, Cairo, Egypt

- Led UI development for the Profiler product and delivered core visualization features adopted by major clients including MediaTek, Microsoft, ARM, and Samsung.
- Engineered a lightweight HTTP server for a desktop profiling tool, enabling integration with verification management solutions and MCP workflows while achieving over 6x faster localhost performance.
- *[Initiative]* Built an agentic issue-resolution workflow that monitors newly opened issues, classifies tasks that can be automated, spawns a custom coding agent to implement fixes, requests code review, and notifies code owners with the rationale behind each change.
- Built a Java/Spring Boot integration between VerificationIQ and a C++-based profiling tool, enabling seamless consumption of profiling data within the verification management workflow.
- Architected a Dockerized Jenkins CI/CD pipeline for automated code complexity reporting and Perforce Swarm integration, reducing manual effort and improving development velocity.
- Refactored legacy C++ components to improve modularity, maintainability, and extensibility, enabling faster future integrations.
- Integrated Siemens analytics SDK to capture crashes, feature usage, and navigation patterns, generating actionable product insights.
- Mentored interns through structured task ownership, code reviews, and technical guidance across C++, development workflows, and practical AI tool usage.

Software Development Engineer Intern Jun 2022 – Sep 2022
Valeo, Cairo, Egypt

- Enhanced OpenGL-based visualization for faster and smoother sensor data rendering.
- Developed server-side features for real-time weather data recording during acquisition.

Education

Master of Science (MSc) in Computer Engineering Sep 2024 – Present
Cairo University, Egypt GPA: 3.63/4.00

Bachelor of Science (BSc) in Computer and Communication Engineering Sep 2018 – May 2023
Cairo University, Egypt GPA: 3.83/4.00 Excellent with Honors, Sixth in Batch
Graduation Project: *Object Detection in Adverse Weather Conditions* — Sponsored by Valeo (Grade: A)

Technical Skills

Languages: C++, Python, Java, JavaScript, SQL
Frameworks & Tools: Qt, OpenGL, Spring Boot, PyTorch, Flutter, Node.js, Jenkins, Docker, Git, GitHub, GitLab, Perforce, Jira, Confluence
Platforms: Linux, Windows
Spoken Languages: Arabic (Native), English (Fluent)

Certifications

IBM Generative AI Engineering Professional Certificate (In Progress) — 10-course series covering:

- Develop Generative AI Applications: Get Started (Credential ID: 4QHPI06JJ5A4)
- Build RAG Applications: Get Started (Credential ID: KLM0CTCGRD0J)
- Vector Databases for RAG, Advanced RAG with Vector Databases and Retrievers
- Build Multimodal Generative AI Applications
- Fundamentals of Building AI Agents, Agentic AI with LangChain, LangGraph, CrewAI, AutoGen, and BeeAI
- Build AI Agents using MCP, and RAG/Agentic AI Capstone Project

Key Projects

- **Jisr:** Developed a cross-platform crowdfunding platform with edge-AI integration capabilities.
- **Stock Price Prediction (Transformer Model):** Built an autoregressive transformer in Python/PyTorch for financial time-series forecasting.
- **Graphics Engine:** Developed an ECS-based OpenGL graphics engine in C++ with real-time rendering and an FPS demo.
- **Twitter Clone:** Built a Flutter application replicating core Twitter functionality using Bloc state management.